

ASSIGNMENT NO. 10: Window Functions and Analytical Queries

Questions

Q1) Create a MySQL database named SalesDB and a table named SalesData.

SalesData Table

Column Name	Data Type	Constraint
SaleID	INT	PRIMARY KEY
Salesperson	VARCHAR(50)	NOT NULL
Department	VARCHAR(30)	NOT NULL
SaleDate	DATE	NOT NULL
Amount	DECIMAL(10,2)	NOT NULL

Insert at least 12 sales records belonging to three different departments.

Query-

```
CREATE TABLE SalesData (  
    SaleID INT PRIMARY KEY,  
    Salesperson VARCHAR(50) NOT NULL,  
    Department VARCHAR(30) NOT NULL,  
    SaleDate DATE NOT NULL,  
    Amount DECIMAL(10,2) NOT NULL  
);
```

```
INSERT INTO SalesData  
(SaleID, Salesperson, Department, SaleDate, Amount)  
VALUES  
(1, 'Rahul', 'Electronics', '2026-01-05', 45000.00),  
(2, 'Priya', 'Electronics', '2026-01-12', 32000.00),  
(3, 'Amit', 'Electronics', '2026-02-03', 28000.00),  
(4, 'Neha', 'Electronics', '2026-02-15', 51000.00),  
  
(5, 'Riya', 'Clothing', '2026-01-08', 18000.00),  
(6, 'Karan', 'Clothing', '2026-01-20', 25000.00),  
(7, 'Sneha', 'Clothing', '2026-02-10', 22000.00),  
(8, 'Vikas', 'Clothing', '2026-02-25', 30000.00),  
  
(9, 'Aarav', 'Furniture', '2026-01-10', 55000.00),  
(10, 'Anjali', 'Furniture', '2026-01-18', 42000.00),  
(11, 'Rohit', 'Furniture', '2026-02-12', 38000.00),  
(12, 'Isha', 'Furniture', '2026-02-28', 60000.00);
```

Output-

Server: 127.0.0.1 » Database: collegedb » Table: SalesData

Showing rows 0 - 11 (12 total, Query took 0.0015 seconds.)

`SELECT * FROM SalesData;`

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Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

	SaleID	Salesperson	Department	SaleDate	Amount
☐	1	Rahul	Electronics	2026-01-05	45000.00
☐	2	Priya	Electronics	2026-01-12	32000.00
☐	3	Amit	Electronics	2026-02-03	28000.00
☐	4	Neha	Electronics	2026-02-15	51000.00
☐	5	Riya	Clothing	2026-01-08	18000.00
☐	6	Karan	Clothing	2026-01-20	25000.00
☐	7	Sneha	Clothing	2026-02-10	22000.00
☐	8	Vikas	Clothing	2026-02-25	30000.00
☐	9	Aarav	Furniture	2026-01-10	55000.00
☐	10	Anjali	Furniture	2026-01-18	42000.00
☐	11	Rohit	Furniture	2026-02-12	38000.00
☐	12	Isha	Furniture	2026-02-28	60000.00

Q2) Write a MySQL query using ROW_NUMBER() to assign a unique row number to every salesperson.

The result should be arranged by sale amount from highest to lowest.

Required Output

Column
Salesperson
Department
Amount
SalesRowNumber

Query-

```
SELECT
Salesperson,
Department,
Amount,
ROW_NUMBER() OVER(
ORDER BY Amount DESC, SaleID ASC
) AS SalesRowNumber
FROM SalesData;
```

Output-

✓ Showing rows 0 - 11 (12 total, Query took 0.0010 seconds.)

```
SELECT Salesperson, Department, Amount, ROW_NUMBER() OVER ( ORDER BY Amount DESC, SaleID ASC ) AS SalesRowNumber FROM SalesData;
```

☐ Profiling [[Edit inline](#)] [[Edit](#)] [[Explain SQL](#)] [[Create PHP code](#)] [[Refresh](#)]

☐ Show all | Number of rows: 25 ▼ Filter rows: Sort by key: None ▼

Extra options

Salesperson	Department	Amount	SalesRowNumber
Isha	Furniture	60000.00	1
Aarav	Furniture	55000.00	2
Neha	Electronics	51000.00	3
Rahul	Electronics	45000.00	4
Anjali	Furniture	42000.00	5
Rohit	Furniture	38000.00	6
Priya	Electronics	32000.00	7
Vikas	Clothing	30000.00	8
Amit	Electronics	28000.00	9
Karan	Clothing	25000.00	10
Sneha	Clothing	22000.00	11
Riya	Clothing	18000.00	12

Q3) Write a MySQL query using RANK() to rank salespersons according to their sale amount.

Students having the same sale amount should receive the same rank.

Required Output

Column
Salesperson
Department
Amount
SalesRank

Query-

```
SELECT  
Salesperson,  
Department,  
Amount,  
RANK() OVER (  
    ORDER BY Amount DESC  
) AS SalesRank  
FROM SalesData;
```

Output-

✓ Showing rows 0 - 11 (12 total, Query took 0.0006 seconds.)			
<pre>SELECT Salesperson, Department, Amount, RANK() OVER (ORDER BY Amount DESC) AS SalesRank FROM SalesData;</pre>			
☐ Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]			
☐ Show all	Number of rows: 25 ▼	Filter rows: Search this table	Sort by key: None ▼
Extra options			
Salesperson	Department	Amount	SalesRank
Isha	Furniture	60000.00	1
Aarav	Furniture	55000.00	2
Neha	Electronics	51000.00	3
Rahul	Electronics	45000.00	4
Anjali	Furniture	42000.00	5
Rohit	Furniture	38000.00	6
Priya	Electronics	32000.00	7
Vikas	Clothing	30000.00	8
Amit	Electronics	28000.00	9
Karan	Clothing	25000.00	10
Sneha	Clothing	22000.00	11
Riya	Clothing	18000.00	12

Q4) Write a MySQL query to rank salespersons separately within each department.

Use:

Required Component
RANK()
OVER
PARTITION BY Department
ORDER BY Amount DESC

Query-

```
SELECT
Salesperson,
Department,
Amount,
RANK() OVER (
    PARTITION BY Department
    ORDER BY Amount DESC
) AS DepartmentRank
FROM SalesData
ORDER BY Department, DepartmentRank;
```

Output-

✓ Showing rows 0 - 11 (12 total, Query took 0.0006 seconds.) [Department: **CLOTHING...** - **FURNITURE...**]

`SELECT Salesperson, Department, Amount, RANK() OVER ( PARTITION BY Department ORDER BY Amount DESC ) AS DepartmentRank FROM SalesData ORDER BY Department, DepartmentRank;`

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☐ Show all | Number of rows: | Filter rows: | Sort by key:

Extra options

Salesperson	Department	Amount	DepartmentRank
Vikas	Clothing	30000.00	1
Karan	Clothing	25000.00	2
Sneha	Clothing	22000.00	3
Riya	Clothing	18000.00	4
Neha	Electronics	51000.00	1
Rahul	Electronics	45000.00	2
Priya	Electronics	32000.00	3
Amit	Electronics	28000.00	4
Isha	Furniture	60000.00	1
Aarav	Furniture	55000.00	2
Anjali	Furniture	42000.00	3
Rohit	Furniture	38000.00	4

Q5) Write a MySQL query to calculate the running total of sales according to the sale date.

Required Output

Column
SaleID
Salesperson
SaleDate
Amount
RunningTotal

Query-

```
SELECT
SaleID,
Salesperson,
SaleDate,
Amount,
SUM (Amount) OVER (
    ORDER BY SaleDate, SaleID
    ROWS BETWEEN UNBOUNDED PRECEDING AND CURRENT ROW
) AS RunningTotal
FROM SalesData
ORDER BY SaleDate, SaleID;
```

Output-

Showing rows 0 - 11 (12 total, Query took 0.0008 seconds) [SaleDate: 2026-01-05... - 2026-02-28...] [SaleID: 1... - 12...]

```
SELECT SaleID, Salesperson, SaleDate, Amount, SUM(Amount) OVER ( ORDER BY SaleDate, SaleID ROWS BETWEEN UNBOUNDED PRECEDING AND CURRENT ROW ) AS RunningTotal FROM SalesData ORDER BY SaleDate, SaleID;
```

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☐ Show all | Number of rows: 25 | Filter rows: Search this table

Extra options

	SaleID	Salesperson	SaleDate	Amount	RunningTotal
☐ Edit Copy Delete	1	Rahul	2026-01-05	45000.00	45000.00
☐ Edit Copy Delete	5	Riya	2026-01-08	18000.00	63000.00
☐ Edit Copy Delete	9	Aarav	2026-01-10	55000.00	118000.00
☐ Edit Copy Delete	2	Priya	2026-01-12	32000.00	150000.00
☐ Edit Copy Delete	10	Anjali	2026-01-18	42000.00	192000.00
☐ Edit Copy Delete	6	Karan	2026-01-20	25000.00	217000.00
☐ Edit Copy Delete	3	Amit	2026-02-03	28000.00	245000.00
☐ Edit Copy Delete	7	Sneha	2026-02-10	22000.00	267000.00
☐ Edit Copy Delete	11	Rohit	2026-02-12	38000.00	305000.00
☐ Edit Copy Delete	4	Neha	2026-02-15	51000.00	356000.00
☐ Edit Copy Delete	8	Vikas	2026-02-25	30000.00	386000.00
☐ Edit Copy Delete	12	Isha	2026-02-28	60000.00	446000.00

Q6) Write a MySQL query to display the highest-performing salesperson from each department.

The query should:

Requirement
Use ROW_NUMBER()
Partition records by department
Arrange sale amounts in descending order
Use a subquery
Display only the first-ranked salesperson from every department

Query-

```
SELECT Salesperson, Department, Amount
FROM (
  SELECT
    Salesperson,
    Department,
    Amount,
    ROW_NUMBER() OVER (
      PARTITION BY Department
      ORDER BY Amount DESC
    ) AS RowNum
  FROM SalesData
) AS RankedSales
WHERE RowNum = 1;
```

Output-

Showing rows 0 - 2 (3 total, Query took 0.0011 seconds)

SELECT Salesperson, Department, Amount FROM (SELECT Salesperson, Department, Amount, ROW_NUMBER() OVER (PARTITION BY Department ORDER BY Amount DESC) AS RowNum FROM SalesData) AS RankedSales WHERE RowNum = 1;

Profiling

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Create PHP code

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Show all

Number of rows: 25

Filter rows: Search this table

Extra options

Salesperson	Department	Amount
Vikas	Clothing	30000.00
Neha	Electronics	51000.00
Isha	Furniture	60000.00